Experiment 9
Operational Amplifier Circuits

Components/Instruments/Meters

- Digital storage oscilloscope
- Function generator
- DC power supply
- Breadboard
- Resistors
- 741 IC
- Connecting wires

Brief description

The operational amplifier is an electronic circuit component that can be used to perform mathematical operations, such as addition, subtraction, scaling, integration, differentiation etc., on voltage and currents. In this experiment, you will be studying the basic functionality of an op-amp (short for operational amplifier) and learn how to use it to perform amplification and addition.

LM741 OpAmp IC

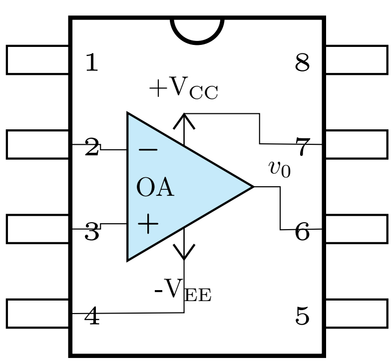


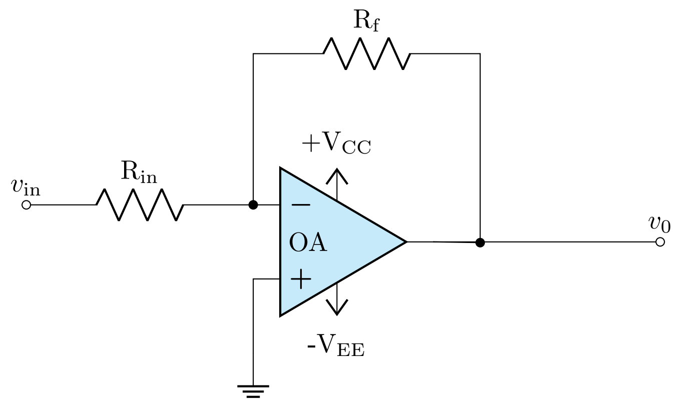
Figure 1: Pin diagram of LM741 IC

Table 1: LM741 IC connections

Pin number	Connection
1	NA
2	Inverting input
3	Non-inverting input
4	-12 V
5	NA
6	Output
7	+ 12 V
8	NA

Part 1: Inverting amplifier

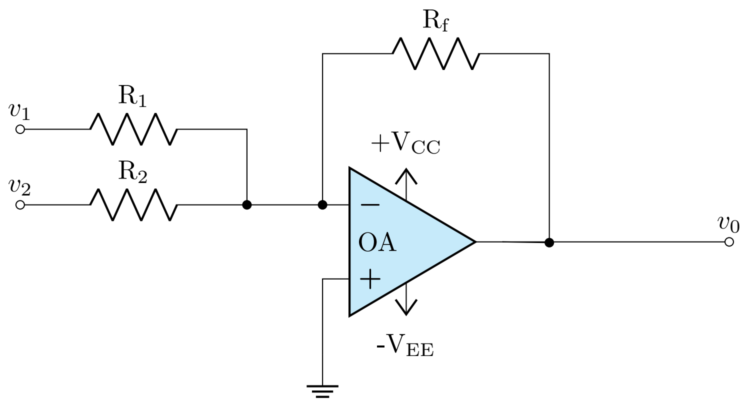
- Hook up the circuit as shown below



- Start with any values of resistances between 1 kΩ and 10 kΩ.
- Set the signal generator to "Square wave" signal (Peak-to-peak value: V_{pp}=1.0 V, frequency= 500 Hz∼1000 Hz).
- Connect 'Channel-1' of DSO to input terminals of the circuit. Press "Auto-scale" button and thereafter adjust the time base properly so that 3∼5 cycles of input signal are displayed on the DSO screen.
- Connect 'Channel-2' of DSO to output terminals of the circuit, i.e. output voltage of the "Inverting Amplifier Circuit".
- Keeping R_{in} constant, figure out the value of R_f at which the peak to peak value of the output voltage is twice that of the input.
- Now, adjust R_{in} such that the peak to peak value of the output voltage is thrice that of the input.

Part 2: Summer

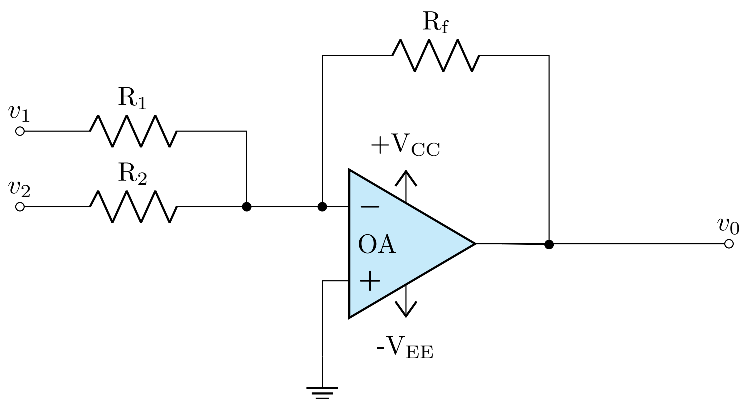
- Hook up the circuit as shown below



- Take $R_1 = R_2 = R_f = 1$ kΩ.
- Set the function generator as follows:
Channel-1: Peak-to-peak value = 1 V, frequency = 100 Hz
Channel-2: Peak-to-peak value = 1 V, frequency = 100 Hz
- Observe the output on the DSO, and work out the relationship between the output and the two inputs.

Part 3: Binary to Decimal Converter (Summer + Amplifier)

- Hook up the circuit as shown below



- Set the function generator to "Square wave" as follows:
  Channel-1: Peak-to-peak value = 1 V, frequency = 100 Hz
  Channel-2: Peak-to-peak value = 1 V, frequency = 100 Hz

- Design a circuit (identify R_1, R_2 and R_f) that produces the following output:

Channel 1 (V_{pp})	Channel 2 (V_{pp})	Output (V_{pp})
0	0	0
1	0	1
0	1	2
1	1	3

- How would your circuit change if the desired output is

Channel 1 (V_{pp})	Channel 2 (V_{pp})	Output (V_{pp})
0	0	0
1	0	2
0	1	4
1	1	6

- Verify your results on the DSO.